

Impact Statement

A Sesotho-English Agricultural Advisory Small Language Model for Smallholder Farmers in Lesotho

Purpose and Context

This project develops a domain-specific small language model that delivers verified, locally grounded agricultural advice to smallholder farmers in Lesotho through a lightweight, offline-first smartphone text app in Sesotho and English. The anticipated impacts below cover societal, ethical, and economic effects as they may unfold over the short term (1-2 years) and longer term (3-5 years).

Positive Societal Impacts

Food security and household income. Timely, crop-specific planting and pest advice addresses the decision failures our problem statement identifies; even modest improvements in planting timing can protect a full season's yield for a household that depends on it.

Linguistic dignity. A model that answers in fluent, locally validated Sesotho affirms the language's value in a technical domain and preserves agricultural vocabulary and indigenous knowledge (e.g., traditional planting indicators) in digital form.

Extension capacity. With few extension officers in Lesotho, the model serves as a triage layer, answering routine queries so officers can focus on complex cases — multiplying the reach of the public extension service.

Closing the information gap. The dataset deliberately includes women farmers, older farmers, remote mountain districts, and non-smartphone users via the extension-officer channel, directing benefits toward those currently most excluded.

Risks and Ethical Considerations

Harm from wrong advice. An incorrect recommendation can cost a household its season. Mitigations include verified-source grounding, confidence indicators, an "ask an extension officer" class in training data, and an SMS grievance channel. Residual risk remains, especially as climate patterns shift beyond historical data.

Automation complacency. Over time, farmers or officers may defer to the model over local expertise. We position the model as a complement to, never a replacement for, extension officers, and source citation is mandatory in every answer.

Digital divide. If the app assumes smartphone ownership, it could widen the gap it aims to close. Offline-first design and officer-mediated access mitigate this, but we acknowledge we cannot reach every farmer.

Data exploitation. Community knowledge could be extracted without lasting benefit. Farmers and extension officers sit on the dataset review board, consent (oral, in Sesotho) is revocable, and synthetic data is flagged so authentic community voices are never misrepresented.

Impact on Core Values

Fairness is addressed through quota-based representation across agro-ecological zones, genders, and age groups, audited in a published distribution report with each release. Dignity and autonomy are respected through consent in the user's language, the right to withdraw data, and cited sources that let farmers judge advice themselves. Community well-being is the end goal; accountability is maintained through a community advisory panel, a public Data Card, and annual findings reported back to districts in Sesotho.

Uncertainties and Limitations

Uptake among farmers with low digital literacy is uncertain. Climate change may erode the validity of historical planting data and will require continuous updating. Benefits depend on sustained partnerships with the Ministry of Agriculture and Meteorological Services, which are outside our control. Most importantly, impact is measured in the field, not the lab: our success metrics (usability, trust, narrowing of the information gap) require multi-season community evaluation before strong claims can be made.